

19. (3 points)

Given the following financial projections for an insurer as of December 31, 2016 (\$000,000):

	<u>Calendar Year</u>		
	<u>2017</u>	<u>2018</u>	<u>2019</u>
Beginning US GAAP equity	1,000	xxx	xxx
Projected Net Income	80	100	140
Minimum Capital to maintain AA Rating at year end	1,015	1,035	1,040
Minimum Capital required by governing regulator at year end	813	840	875
Minimum Capital to meet management's growth target at year end	1,017	1,015	1,035
Change in loss and expense reserves	100	-75	25
Net borrowing	12	0	15

- Risk-free rate: 2.0%
- Expected equity market risk premium: 6.0%
- Insurance company equity beta: 1.25

a. (0.5 point)

Determine the required equity return percentage based on the Capital Asset Pricing Model (CAPM).

b. (2.5 points)

Determine the value of this company as of January 1, 2017 based on the Free Cash Flow to Equity (FCFE) method.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 19

TOTAL POINT VALUE: 3

LEARNING OBJECTIVE: B1: Calculate the effect of loss and expense reserve requirements and regulatory or rating agency capital requirements on the free cash flow to equity for a P&C insurer.

B2: Value the equity of a P&C insurer based on its expected future dividends, its free cash flow to equity, or its expected abnormal earnings.

SAMPLE ANSWERS

Part a: 0.5 point

Sample Answer 1

$$k = r_f + \beta (E(r_m) - r_f) = 0.02 + 1.25 \times (0.06) = 0.95$$

Sample Answer 2

$$k = 0.02 + 1.25 \times (0.06) = 0.95$$

Part b: 2.5 points

Sample Answer 1

Use highest capital requirement for each year

Loss reserves cancel out in FCFE equation – don't include

$$FCFE = NI + \text{Net Borrowing} + \text{Non-Cash Charges} - \Delta \text{ capital} - \Delta \text{ WC}$$

CY	2017	2018	2019
NI	80	100	140
Beg Equity	1000	1017	1035
Ending Eq	1017	1035	1040
Net Borrow	12	0	15
↑ Cap	17	18	5
FCFE	75	82	150
ROE	0.08	0.098	0.135
Reinv	0.2125	0.18	0.036
Growth	0.017	0.0176	0.0048

Since there is no trend in growth and it's volatile use an avg = 0.013

$$\frac{75}{1.095} + \frac{82}{1.095^2} + \frac{150}{1.095^3} + \frac{150(1.013)}{0.095 - 0.013} = 251.13 + 1411.38 = 1662.51M$$

Sample Answer 2

FCFE method

FCFE = Net Income + (graders note: the written out formula was incomplete, but used correctly and given full credit)

Selected Max(Min Cap AA Rating, Min Cap regulator, Min Cap Growth)

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

	2017	2018	2019	2020
Beginning Equity	1000	1017	1035	1040
NI	80	100	140	
Δ Capital	17	18	5	
Net Borrow	12	0	15	
FCFE	75	82	150	

$$PVFH = \frac{75}{1.095} + \frac{82}{1.095^2} + \frac{150}{1.095^3} = 251.13$$

Calculate Growth	2017	2018	2019	Selected	
ROE = NI/RE	.08	.098	.135	13%	← going with close to CY 19, since trending upward
Reinvest = ΔCap/NI	.2125	.18	.036	12.5%	← weighted average, also trending downward which takes that into account without going all the way to 4% (if anomaly)

$$G = ROE \times \text{reinvest rate} = 13\% \times 12.5\% = .01625$$

$$TV = \frac{150(1.01625)}{.095 - .01625} = 1935.714$$

$$PVTV = \frac{1935.714}{1.095^3} = 1474.34$$

$$\text{Value of Company} = 251.14 + 1474.34 = 1725.47 \text{ (in 000,000)}$$

EXAMINER'S REPORT

- On this question, the candidate was expected to know how to value the equity of a P&C insurer based on its free cash flow to equity
- Candidates generally scored well, either gaining full credit or taking slight deductions for minor mistakes/lack of explanation.
- Generally speaking, the candidates understood what was asked of them. Many, however, lost partial credit for various mistake (see part b for details)

Part a

- Candidates were expected to know how to calculate the required return for the firm.

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

- While mistakes weren't common, of the responses that were incorrect, the most common error was to use the market risk premium as the expected market return.

Part b

- Candidates were expected to know how to calculate the value of the firm through free cash flow to equity.
- Most candidates appeared familiar with how to go about addressing this problem. Mistakes were mostly in details such as:
 - Choice of equity at the end of each year
 - Inclusion/exclusion of net borrowing in the FCFE calculation
 - Inclusion/exclusion of reserves in the FCFE calculation
 - To calculate the growth rate, relying on a formula that was specific to one example in the textbook: $\text{Reinvested Rate} = (\text{Net Income} - \text{Free Cash Flow}) / \text{Net Income}$. This example did not include a debt component.

As there is a debt component in this item, the formula to use is $\text{Reinvested Rate} = (\text{Net Income} - \text{Free Cash Flow} + \Delta \text{Debt}) / \text{Net Income}$, or $\text{Reinvested Rate} = \Delta \text{capital} / \text{Net Income}$. Alternatively, growth rate could have been calculated directly as $\text{ending equity} / \text{beginning equity} - 1$.
- The terminal value and the discount factors were generally calculated correctly
- Some candidate lost partial credit for a lack of explanation or for not clearly stating assumptions for the following
 - Choice of equity at the end of the year
 - Choice of growth rate if selection was based on some multi-year average and/or excluded an "abnormal" year